



Precise genomic deletions using paired prime editing

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Current methods to delete genomic sequences are based on clustered regularly interspaced short palindromic repeats (CRISPR)–Cas9 and pairs of single-guide RNAs (sgRNAs), but can be inefficient and imprecise, with errors including small indels as well as unintended large deletions and more complex rearrangements. In the present study, we describe a prime editing-based method, PRIME-Del, which induces a deletion using a pair of prime editing sgRNAs (pegRNAs) that target opposite DNA strands, programming not only the sites that are nicked but also the outcome of the repair. PRIME-Del achieves markedly higher precision than CRISPR–Cas9 and sgRNA pairs in programming deletions up to 10 kb, with 1–30% editing efficiency. PRIME-Del can also be used to couple genomic deletions with short insertions, enabling deletions with junctions that do not fall at protospacer-adjacent motif sites. Finally, extended expression of prime editing components can substantially enhance efficiency without compromising precision. We anticipate that PRIME-Del will be broadly useful for precise, flexible programming of genomic deletions, epitope tagging and, potentially, programming genomic rearrangements.

The ability to precisely manipulate the genome can critically enable investigations of the function of specific genomic sequences, including genes and regulatory elements. Within the past decade, CRISPR–Cas9-based technologies have proved transformative in this regard, allowing precise targeting of a genomic locus, with a quickly expanding repertoire of editing or perturbation modalities¹. Among these, the precise and unrestricted deletion of specific genomic sequences is particularly important, with critical use cases in both functional genomics and gene therapy.

Currently, the leading method for programming genomic deletions uses a pair of CRISPR sgRNAs that each target a protospacer-adjacent motif (PAM) sequence, generating a pair of nearby DNA double-strand breaks (DSBs). On simultaneous cutting of two sites, cellular DNA damage repair factors often ligate two ends of the genome without the intervening sequence² through nonhomologous end-joining (NHEJ) (Fig. 1a). Although powerful, this approach has several limitations: (1) an attempt to induce a deletion, particularly a longer deletion, often results in short insertions or deletions (indels; typically <10 bp) near one or both DSBs, with or without the intended deletion^{3–5}; (2) other unintended mutations, including large deletions and more complex rearrangements, can frequently occur and go undetected for technical reasons^{5–8}; (3) DSBs are a cytotoxic insult⁹; and (4) the junctions of genomic deletions programmed by this method are limited by the distribution of naturally occurring PAM sites. Notwithstanding these limitations, various studies have employed this strategy to great effect, for example, to investigate the function of genes and regulatory elements^{5,10,11}, as well as toward gene therapy^{12,13}. However, limited precision, DSB toxicity and the inability to program arbitrary deletions have handicapped the utility of CRISPR–Cas9-induced deletions in functional and therapeutic genomics.

Recently, Liu and colleagues described ‘prime editing’, which expands the CRISPR–Cas9 genome editing toolkit in critical ways¹⁴.

Prime editing utilizes a Prime Editor-2 enzyme, which is a Cas9 nickase (Cas9 H840A) fused with a reverse transcriptase, and a 3′-extended sgRNA (pegRNA). The Prime Editor-2 enzyme and pegRNA complex can nick one strand of the genome and attach a 3′-single-stranded DNA flap to the nicked site, following the template RNA sequence in the pegRNA molecule. By including homologous sequences in the neighboring region, DNA damage repair factors can incorporate the 3′-flap sequence into the genome. The incorporation rate can be further enhanced using an additional sgRNA, which makes a nick on the opposite strand, boosting DNA repair with the 3′-flap sequence but often with a decrease in precision (strategy referred to as PE3/PE3b)¹⁴ (Fig. 1b). The principal advantage of prime editing lies with its encoding of both the site to be targeted and the nature of the repair within a single molecule, the pegRNA. In addition to demonstrating many other classes of precise edits, Anzalone et al.¹⁴ used the PE3 strategy to show that a single pegRNA/sgRNA pair could be used to program deletions ranging from 5 bp to 80 bp, achieving high efficiency (52–78%) with modest precision (on average, 11% rate of unintended indels)¹⁴. However, even the PE3 strategy could face difficulties in programming deletions >100 bp, because, at least in plants, observed efficiencies fall precipitously for deletions >20 bp (ref. 15).

We reasoned that a pair of pegRNAs could be used to specify not only the sites that are nicked but also the outcome of the repair, potentially enabling programming of deletions longer than 100 bp (Fig. 1c). In the present study, we demonstrate that this strategy, which we call PRIME-Del, induces the efficient deletion of sequences up to 10 kb in length with much higher precision than observed or expected with either the Cas9/paired-sgRNA or PE3 strategies. We furthermore show that PRIME-Del can concurrently program short insertions at the deletion site. Concurrent deletion/insertion can be used to introduce in-frame deletions, to introduce epitope tags concurrently with deletions and, more generally, to

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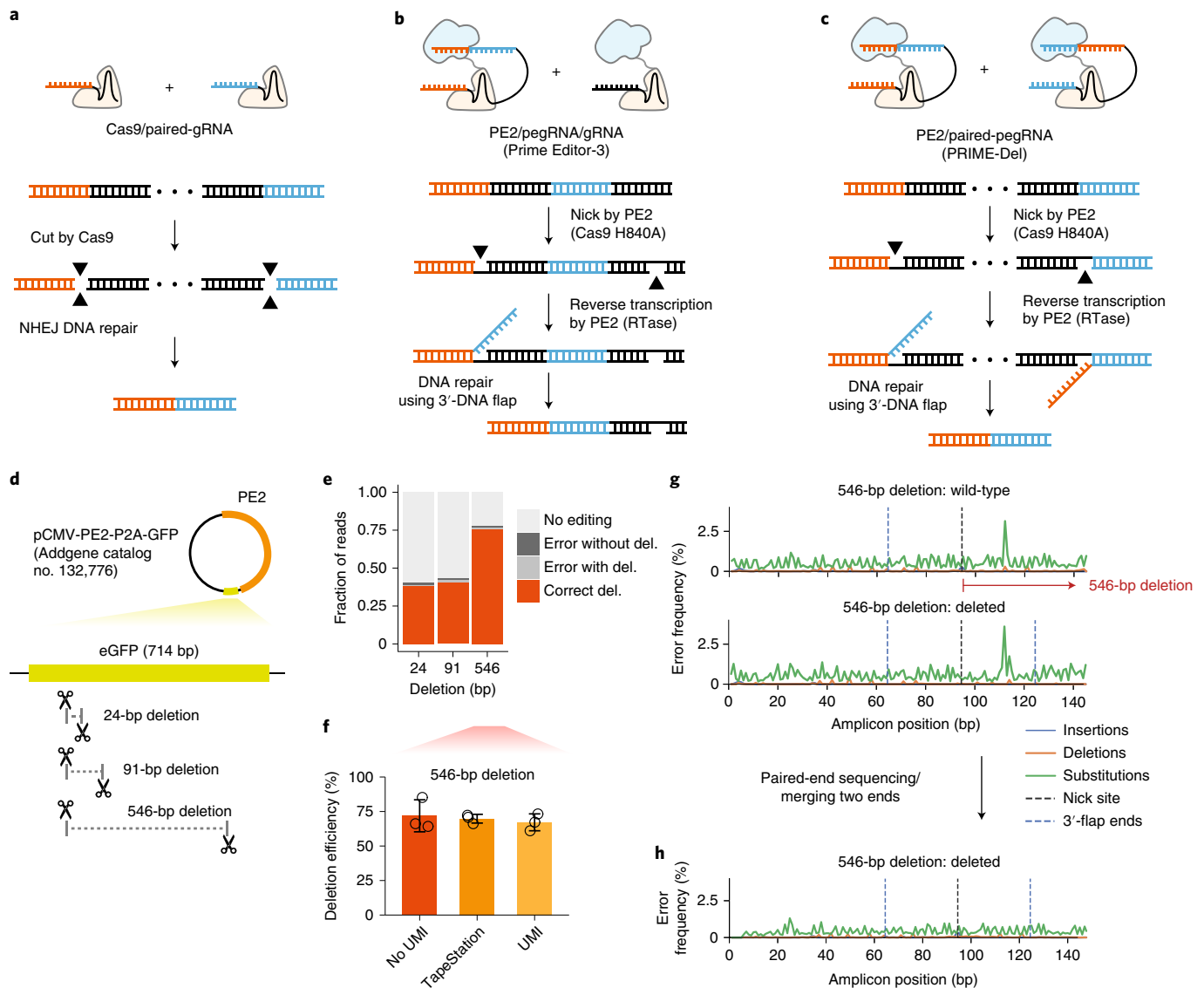


Fig. 1 | Precise episomal deletions using PRIME-Del. **a–c**, Schematic of Cas9/paired-sgRNA deletion strategy (**a**), PE3 strategy (**b**) and PRIME-Del (**c**). For PRIME-Del, a pair of pegRNAs encodes the sites to be nicked at each end of the intended deletion, as well as a 3'-flap that is complementary to the region targeted by the other pegRNA. **d**, Cartoon representation of deletions programmed within the episomally encoded *eGFP* gene (not drawn to scale). **e**, PRIME-Del-mediated deletion efficiencies and error frequencies (with or without intended deletion) measured for 24-bp, 91-bp and 546-bp deletion experiments in HEK293T cells (mean over $n=5$ transfection replicates). Sequencing reads were classified as being without indel modifications ('No editing'), indel errors without the intended deletion, indel errors with the intended deletion and correct deletion without error. **f**, PRIME-Del-mediated deletion efficiency was measured for the 546-bp deletion experiment using three methods (mean $\pm$ s.d. over $n=3$ transfection replicates). **g**, Insertion, deletion and substitution error frequencies across sequencing reads from 546-bp deletion experiment. Reads were aligned to reference sequence either without (top) or with (bottom) deletion. Plots are from single-end reads with collapse of UMIs to reduce sequencing errors; also shown are additional replicates and error-class-specific scales in Supplementary Fig. 1e. Note that only one of the two 3'-DNA flaps is covered by the sequencing read in amplicons lacking the deletion (labeled as 'wild-type'). **h**, Insertion, deletion and substitution error frequencies across the amplicons from 546-bp deletion experiment after merging paired-end sequencing reads.

facilitate the programming of deletions unrestricted by the endogenous distribution of PAM sites. By filling these gaps, PRIME-Del expands our toolkit to investigate the biological function of genomic sequences at single-nucleotide resolution.

Results and discussion

PRIME-Del induces precise deletions in episomal DNA. We first tested the feasibility of the PRIME-Del strategy by programming deletions to an episomally encoded *eGFP* gene. We designed pairs of

pegRNAs specifying 24-, 91- and 546-bp deletions within the *eGFP* coding region of the pCMV-PE2-P2A-GFP plasmid (Addgene, catalog no. 132776; Fig. 1d). We cloned each pair of pegRNAs into a single plasmid with separate promoters, the human U6 and H1 sequences⁵. We transfected HEK293T cells with *eGFP*-targeting paired-pegRNA and pCMV-PE2-P2A-GFP plasmids. We harvested DNA (including both genomic DNA and residual plasmid) from cells 4–5 d after transfection and PCR amplified the *eGFP* region. We then sequenced PCR amplicons to quantify the efficiency of the

programmed deletion, as well as to detect unintended edits to the targeted sequence.

We calculated deletion efficiency as the number of reads aligning to a reference sequence of the intended deletion, out of the total number of reads aligning to reference sequences either with or without the deletion. Estimated deletion efficiencies ranged from 38% (24-bp deletion) to 77% (546-bp deletion), and were consistent across replicates (note: throughout the paper, the term ‘replicate’ is used to refer to independent transfections) (Fig. 1e). This result clearly indicates that the PRIME-Del strategy outlined in Fig. 1c can work. However, we were initially concerned that these were overestimates of efficiency due to the shorter, edited templates being favored by both PCR and Illumina-based sequencing, particularly for the 546-bp deletion, because it has the largest difference between amplicon sizes (766bp versus 220bp for wild-type and deletion amplicons, respectively). To address this, we repeated the amplification on DNA from the 546-bp deletion experiment with a two-step PCR, first adding 15bp of unique molecular identifiers (UMIs) via linear amplification before a second, exponential phase. The addition of UMIs via linear PCR was intended to minimize PCR and sequencing biases in our estimates of deletion efficiencies¹⁶. PRIME-Del efficiency was assessed based on the sequencing data after collapsing reads with identical UMIs, as well as on the product size distribution (Agilent TapeStation). We observed a slight decrease in deletion efficiency after duplicate removal, from 73% to 66%, comparable to the 70% efficiency measured on the TapeStation (Fig. 1f). These results suggest that our initial estimates of efficiency are only modestly impacted by size-dependent biases.

For most of these sequencing data, we had only a single read extending over the intended deletion site. As such, it was difficult to distinguish unintended editing outcomes (for example, indels at the nick sites) from PCR or sequencing errors. To address this, in part, we plotted frequencies of different classes of errors (substitutions, insertions, deletions) for sequences aligning to either the unedited sequence (Fig. 1g, top) or the intended deletion (Fig. 1g, bottom), along the length of the sequencing read. For all replicates of the three deletion experiments (Supplementary Fig. 1), these profiles showed low rates of substitutions and indels, with almost identical profiles and no consistent increase in the rate of any class of error at either the positions of the Prime Editor-2 enzyme nick sites or 3′-flap ends >1%, particularly after collapse by UMI (Fig. 1g and Supplementary Fig. 1e) or repeating sequencing with longer, paired-end sequencing reads (Fig. 1h).

Simultaneous deletion and short insertion using PRIME-Del. We reasoned that, as the homology sequences in the 3′-flaps program the deletion, we could potentially use PRIME-Del to concurrently introduce a short insertion at the deletion junction (Fig. 2a). The desired insertion would be encoded into the pair of pegRNAs in a reverse complementary manner, just 5′ to the deletion-specifying homology sequences. With the conventional strategy for programming deletions, that is, with Cas9 and paired-sgRNAs, the deletion junctions are determined by the sgRNA targets, the selection of which is limited by the natural distribution of PAM sites (Fig. 2b). Simultaneous deletion and short (<100bps) insertion with PRIME-Del would offer at least three advantages over this conventional strategy. First, an arbitrary insertion of one to three bases could enable a reading frame to be maintained after editing, for example, for deletions intended to remove a protein domain. Second, an arbitrary insertion could be used to effectively move one or both deletion junctions away from the cut sites determined by the PAM, increasing flexibility to program deletions with base-pair precision. Third, insertion of functional sequences at the deletion junction could allow genome editing with PRIME-Del to be coupled to other experimental goals (for example, protein tagging or insertion of a transcriptional start site).

To test this concept, we designed pegRNA pairs encoding five insertions ranging from 3bp to 30bp at the junction of a 546-bp programmed deletion within *eGFP* (Fig. 2c). Although our main objective was to test the effect of insertion length on deletion efficiency, we chose insertion sequences for their importance in molecular biology: the 3-bp insertion sequence generates an in-frame stop codon. The 6-bp insertion sequence includes the start codon with the surrounding Kozak consensus sequence. The 12-bp insertion sequence includes tandem repeats of the m6A post-transcriptional modification consensus sequence of GGACAT¹⁷. The 21-bp insertion sequence includes the T7 RNA polymerase promoter sequence. The 30-bp insertion sequence encodes the in-frame FLAG-tag peptide sequence when translated. The estimated efficiencies for simultaneous short insertion and deletion within the episomal *eGFP* gene in HEK293T cells were comparable to the 546-bp deletion alone, ranging from 83% to 90% for the various programmed insertions (Fig. 2d). Also, insertion, deletion and substitution error rates at deletion junctions and across programmed insertions were comparable to the background error frequencies (Fig. 2e and Supplementary Fig. 2a). As expected, the vast majority (>99%) of reads containing the programmed deletion also contained the insertion (Fig. 2f), indicating that the full lengths of the pair of 3′-DNA flaps generated following the programmed pegRNA sequences specify the repair outcome (Fig. 2a).

PRIME-Del induces precise deletions in genomic DNA.

Encouraged by our initial results on editing episomal DNA, we next tested PRIME-Del on a copy of the *eGFP* gene integrated into the genome. We first generated the polyclonal HEK293T cells that carry the *eGFP* gene by lentiviral transduction, followed by flow sorting to select green fluorescent protein (GFP)-positive cells (Fig. 3a). We then tested the same pairs of pegRNAs encoding concurrent deletions and insertions (546-bp deletion with or without short insertions at the deletion junction) by transfecting pegRNAs and Prime Editor-2 enzyme without enhanced (e)GFP (pCMV-PE2; Addgene catalog no. 132775) to these cells. Although editing efficiencies decreased substantially in comparison to episomal *eGFP* (7–17%; Fig. 3b), we remained unable to detect errors that were clearly associated with editing (Fig. 3c and Supplementary Fig. 2b). Specifically, there was no consistent pattern of error classes above background level accumulating at the nick site or 3′-DNA-flap incorporation sites. Also, as previously, the vast majority of reads with the 546-bp deletion also contained programmed insertions (Supplementary Fig. 2c).

To test PRIME-Del on native genes, we designed two pairs of pegRNAs that respectively specified 118- and 252-bp deletions within exon 1 of *HPRT1* (Fig. 3d). We have previously performed a scanning deletion screen across the *HPRT1* locus using a Cas9/paired-sgRNA strategy⁵. To directly compare PRIME-Del with Cas9/paired-sgRNAs in programming genomic deletions, we attempted the same deletions with the same guides, but substituting Prime Editor-2 enzyme with Cas9 in transfection of HEK293T cells. We quantified the resulting deletion efficiencies using two independent methods: first, we used the aforementioned strategy of appending a 15-bp UMI sequence via a linear PCR step, before the standard PCR and sequencing readout. The resulting sequencing reads are collapsed by shared UMIs to minimize possible biases introduced in the PCR amplification and sequencing cluster-generation steps. Second, we used droplet-digital PCR (ddPCR), which partitions genomic DNA into emulsion droplets before PCR amplification and fluorescence readout of TaqMan probes within each droplet. We designed our probe to bind at the deletion junction, which would generate fluorescence signals specifically in the presence of the deletion. Our design of the reporter probe aims to quantify the precise editing efficiencies, because errors introduced at the deletion junction are less likely to induce efficient binding of the probe during

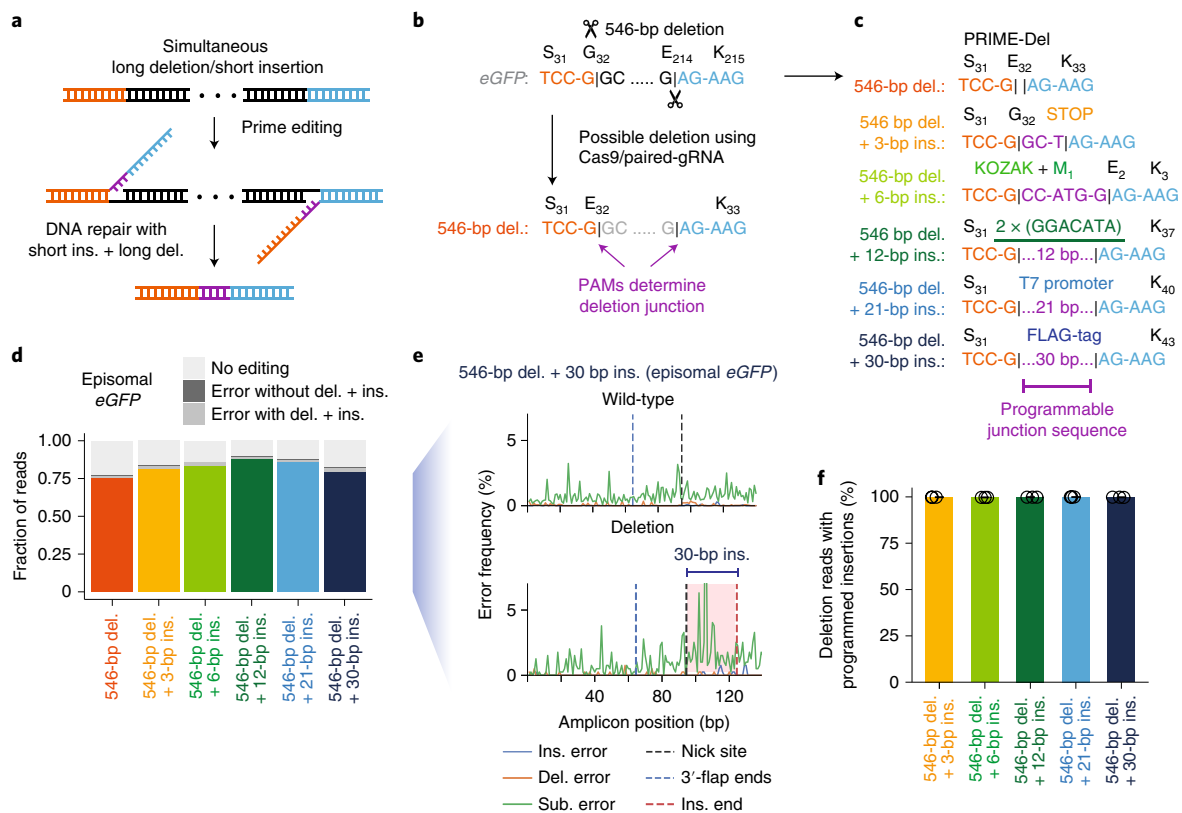


Fig. 2 | Concurrent programming of deletion and insertion using PRIME-Del. a, Schematic of strategy, with reverse complementary sequences corresponding to the intended insertion in purple. **b**, Conventional strategy for deletion with Cas9 and pairs of sgRNAs. Potential deletion junctions are restricted by the natural distribution of PAM sites. **c**, Pairs of pegRNAs designed to encode five insertions (ins.), ranging in size from 3 bp to 30 bp, together with a 546-bp deletion (del.) in *eGFP*. **d**, Estimated deletion efficiencies and indel error frequencies (with or without intended deletion) when using these pegRNA pairs to induce concurrent deletion and insertion in HEK293T cells (mean over $n = 3$ transfection replicates). **e**, Representative insertion, deletion and substitution error frequencies plotted across sequencing reads from the concurrent 546-bp deletion and 30-bp insertion conditions. Plots are from single-end reads without UMI correction. Note that only one of the two 3'-DNA flaps is covered by the sequencing read in amplicons lacking the deletion (labeled as 'wild-type'). **f**, The percentage of reads containing the programmed deletion that also contain the programmed insertion (mean $\pm$ s.d. over $n \geq 3$ transfection replicates).

PCR¹⁸. Signals from deletions were normalized to the reference signal from detecting the copy number of the *RPP30* gene, which has been previously characterized and often used as a standard in ddPCR assays¹⁸. At exon 1 of *HPRT1*, we observed comparable deletion efficiencies for the PRIME-Del and Cas9/paired-sgRNA strategies in HEK293T, ranging from 5% to 30% efficiencies for 118-bp and 252-bp deletions, respectively (Fig. 3e). Of note, we observed consistently lower efficiencies with the ddPCR assay compared with the UMI-based sequencing assay. Although this could be due to overestimation of efficiencies by the UMI-based approach, we also note that PCR amplification of the target region may be inefficient in the ddPCR assay based on the lack of clear separation of fluorescence intensities between positive and negative droplets (Supplementary Fig. 3c,d).

As is well established³⁻⁵, the Cas9/paired-sgRNA strategy often resulted in errors (mostly short deletions), whether with or without the intended deletion (Fig. 3f,g and Supplementary Fig. 3a). Of reads lacking the intended 118-bp or 252-bp deletions, 12% or 12%, respectively, also contained an unintended indel at the observable target site (these are underestimates, because they account for only one of two target sites; Fig. 3f, top). Of reads containing the intended 118-bp or 252-bp deletion, 38% or 34% also contained an unintended indel at the deletion junction, respectively (Fig. 3f, bottom). Such junctional errors are an established consequence of error-prone repair by NHEJ. In contrast, unintended indels were

far less common with PRIME-Del (Fig. 3g and Supplementary Fig. 3b). Of reads lacking the intended 118-bp or 252-bp deletion, 1.1% or 0.5% also contained an unintended short indel at the observable target site, respectively (Fig. 3g, top). Of reads containing the intended 118-bp or 252-bp deletion, 12% or 2.7% also contained an unintended indel at the deletion junction, respectively (Fig. 3g, bottom). The pattern of higher correct editing efficiencies for PRIME-Del over the Cas9/paired-sgRNA strategy is also suggested by the ddPCR measurements, where PRIME-Del reports an almost twofold higher, precisely edited population for both deletions.

For PRIME-Del, especially with the 118-bp deletion on *HPRT1*, the observation of an appreciable rate of insertions at the deletion junction in association with intended deletions (Fig. 3g, bottom, and Supplementary Fig. 3b) contrasts with our earlier observations at *eGFP*, where these rates were consistently equivalent to the background. Further investigation of the error mode revealed that these errors corresponded to long insertions (mean $47 \text{ bp} \pm 12 \text{ bp}$; Supplementary Fig. 4). The most frequent long insertion at the 118-bp deletion junction was 55 bp, a chimeric sequence between two 32-bp 3'-DNA-flap sequences, overlapping at a 'GCCCT' sequence, suggesting its origin from the annealing of GC-rich ends of 3'-DNA flaps. Similar chimeric sequences were observed as insertions at the 252-bp deletion junction, overlapping at 'GCCG' within their 3'-DNA flaps. Nevertheless, even with these long insertions, 82% and 91% of all reads containing an indel matched the intended

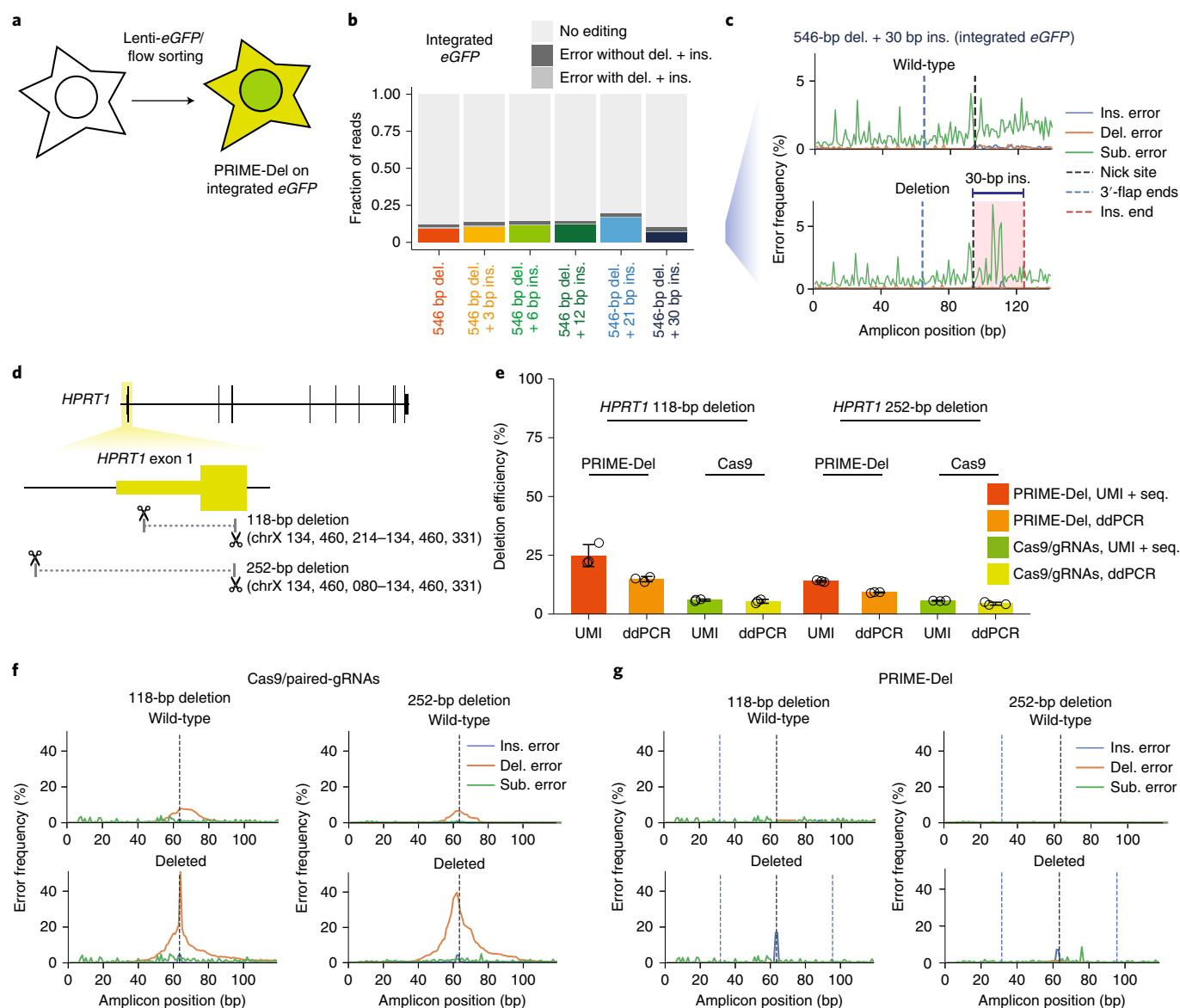


Fig. 3 | Precise genomic deletions using PRIME-Del. **a**, Schematic of generation of the eGFP-integrated HEK293T cell line. **b**, Estimated deletion efficiencies and error frequencies in using PRIME-Del for concurrent deletion and insertion on genomically integrated eGFP in HEK293T cells (mean over $n=3$ transfection replicates). **c**, Representative insertion, deletion and substitution error frequencies plotted across sequencing reads from concurrent 546-bp deletion and 30-bp insertion conditions on genome-integrated eGFP. Plots are from single-end reads without UMI correction. **d**, Cartoon representation of deletions programmed within the *HPRT1* gene. **e**, Deletion efficiencies measured for the 118-bp and 252-bp deletions using either PRIME-Del or Cas9/paired-sgRNA (abbreviated to Cas9) strategies in HEK293T cells, quantified using either the UMI-based sequencing assay or the ddPCR assay (mean $\pm$ s.d. over $n=3$ transfection replicates). **f**, Representative insertion, deletion and substitution error frequencies plotted across sequencing reads from 118-bp deletion (left) and 252-bp deletion (right) at *HPRT* exon 1, using the Cas9/paired-sgRNA strategy. Different error classes are colored the same as in **c**. **g**, Same as **f**, but for PRIME-Del strategy.

deletion exactly with PRIME-Del, but only 38% and 49% with the Cas9/paired-sgRNA strategies (Fig. 4a). Indel errors from the Cas9/paired-sgRNA strategy are probably underestimated, because errors at only one of two Cas9 cut sites are captured by our sequencing strategy.

The structure of the observed insertions and the lack of similar errors in applying PRIME-Del to the eGFP locus suggested that this issue might be addressable through alternative pegRNA designs. As one approach, we either shortened or lengthened the reverse transcription (RT)-template portion of both pegRNAs. For a 118-bp deletion that used 32-bp RT-template lengths for both pegRNAs, we either shortened to 17- and 25-bp-long homology arms or

lengthened to 42- and 46-bp-long homology arms (Supplementary Fig. 5a). Both lengthening and shortening of the homology arms resulted in decreased deletion efficiencies (29% and 26% of the efficiencies observed with the standard designs for short and long homology arms, respectively; Supplementary Fig. 5b). However, among deleted products, lengthening of the homology arms also tended to decrease the long-insertion error frequency (to 30% of the standard design), whereas shortening of the homology arms increased the insertion error frequency (to 129% of the standard design; Supplementary Fig. 5d). Similar trends were observed with the 252-bp deletion, where shortening or lengthening of homology arms decreased the deletion efficiency (Supplementary Fig.

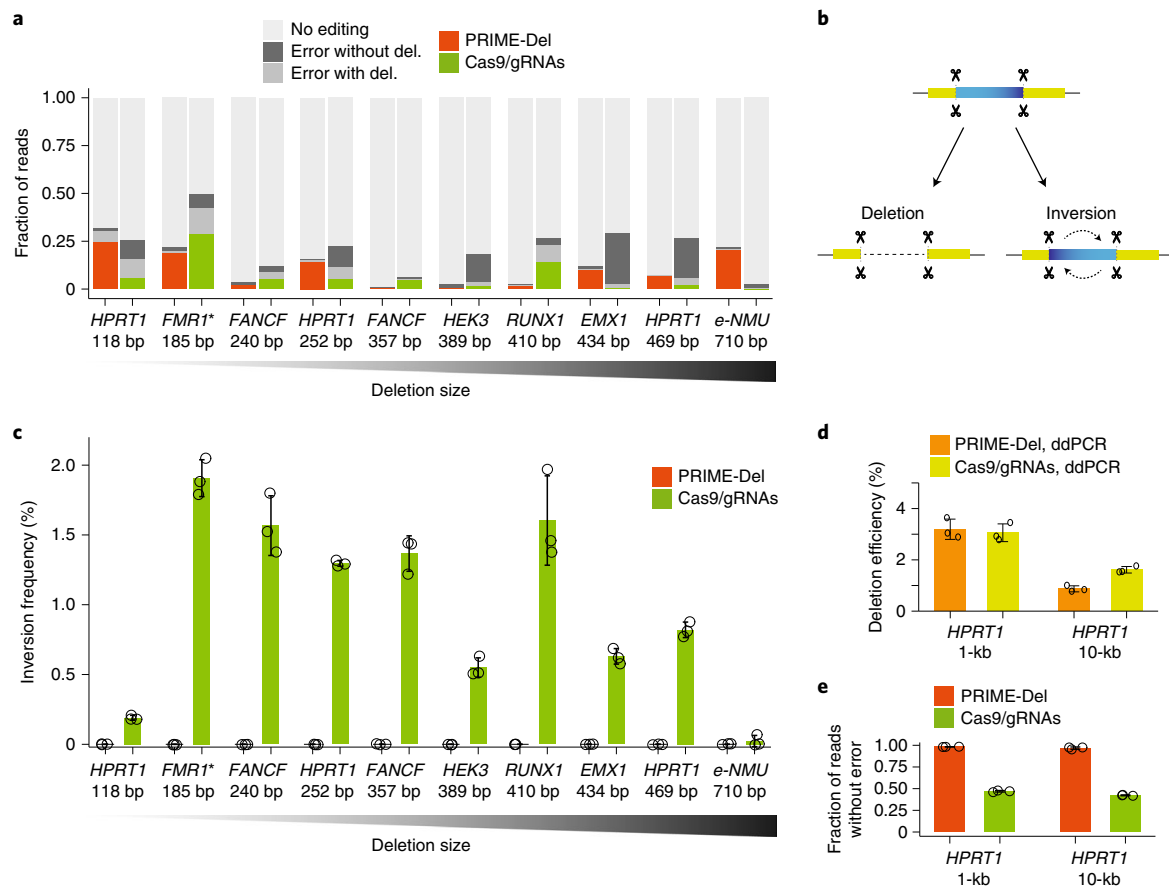


Fig. 4 | Characterizing PRIME-Del across the genome. **a**, Estimated deletion efficiencies and indel error frequencies for different deletions across the genome for PRIME-Del (left) and Cas9/paired-sgRNA (right) methods (mean over $n = 3$ transfection replicates). UMI-based sequencing assay was used for quantification (except the GC-rich amplicon of *FMR1**, where added dimethylsulfoxide interfered with the UMI-addition reaction). **b**, Schematic of a sequence inversion event, which is a known error mode in Cas9/paired-sgRNA-mediated deletion. **c**, Estimated inversion frequencies for different deletions across the genome for PRIME-Del (left) and Cas9/paired-sgRNA (right) methods (mean over $n = 3$ transfection replicates). Note that, whereas they are observed for all but one of the Cas9/paired-sgRNA-mediated deletions at an appreciable frequency, virtually no inversions are observed for any of these ten deletions using PRIME-Del. **d**, Deletion efficiencies measured for 1-kb and 10-kb deletions at *HPRT1* using either PRIME-Del (left) or Cas9/paired-sgRNA (right) with ddPCR-based assay in HEK293T cells (mean $\pm$ s.d. over $n = 3$ transfection replicates). **e**, Fraction of reads with precise deletion measured for the 1-kb and 10-kb deletion on the *HPRT1* gene with either PRIME-Del (left) or Cas9/paired-sgRNA (right), using sequencing of the deletion amplicons (mean $\pm$ s.d. over $n = 3$ transfection replicates).

5c), whereas lengthening of the homology arm increased precision (Supplementary Fig. 5e). As a further control, substituting the sequence of the RT template to that used for programming a 546-bp deletion at *eGFP* failed to induce deletions for both 118-bp and 252-bp constructs targeting *HPRT1* (Supplementary Fig. 5b,c), fortifying the conclusion that PRIME-Del deletions are specific to DNA repair guided by the homology arm sequences.

We further applied genomic deletion using PRIME-Del at additional native loci, altogether testing ten different deletions at seven loci (Fig. 4a). We performed all deletions in HEK293T cells, quantified deletion efficiencies and error frequencies using UMI-based sequencing assay and directly compared PRIME-Del with the Cas9/paired-sgRNA method (that is, using the same guides but substituting in Cas9). Deletion sizes ranged from 118 bp at *HPRT1* exon 1 to 710 bp at *e-NMU* (enhancer for *NMU* gene) locus^{19,20}. In all ten cases, we observed substantially lower error rates with PRIME-Del compared with the Cas9/paired-sgRNA method. In five of ten cases, we observed that the precise deletion is more efficient with PRIME-Del compared with the Cas9/paired-sgRNA method, suggesting that higher precision does not compromise the deletion efficiencies in general. We did not observe a strong relationship

between the deletion size and efficiency in this range (118–710 bp) for either method.

Inversion of the sequence between two DSBs is a well-documented phenomenon when using the Cas9/paired-sgRNA method^{3,21} (Fig. 4b). To understand the frequency of inversion events using PRIME-Del, we aligned sequencing reads to a reference that was generated by inverting the sequence between two nick sites. Across ten deletions in seven loci at which we performed PRIME-Del, we observed that virtually no reads aligned to the inverted reference (Fig. 4c), whereas, for Cas9/paired-sgRNA controls, inversions were detected in up to 2% of reads (Fig. 4c).

To evaluate the length limits of PRIME-Del, we designed two additional deletions, sized 1,064 bp (1 kb) and 10,204 bp (10 kb) at the *HPRT1* locus. As our sequencing-based assay is not well suited to detect amplicons >1 kb, we used sequencing to quantify error frequencies in the deletion product alone, and ddPCR to measure the efficiency of precise deletion, again comparing Prime Editor-2 and Cas9 side by side. We observed that, although deletion efficiencies between PRIME-Del and the Cas9/paired-sgRNA method were comparable in HEK293T cells (Fig. 4d), PRIME-Del achieves much higher precision, consistent with our observations while

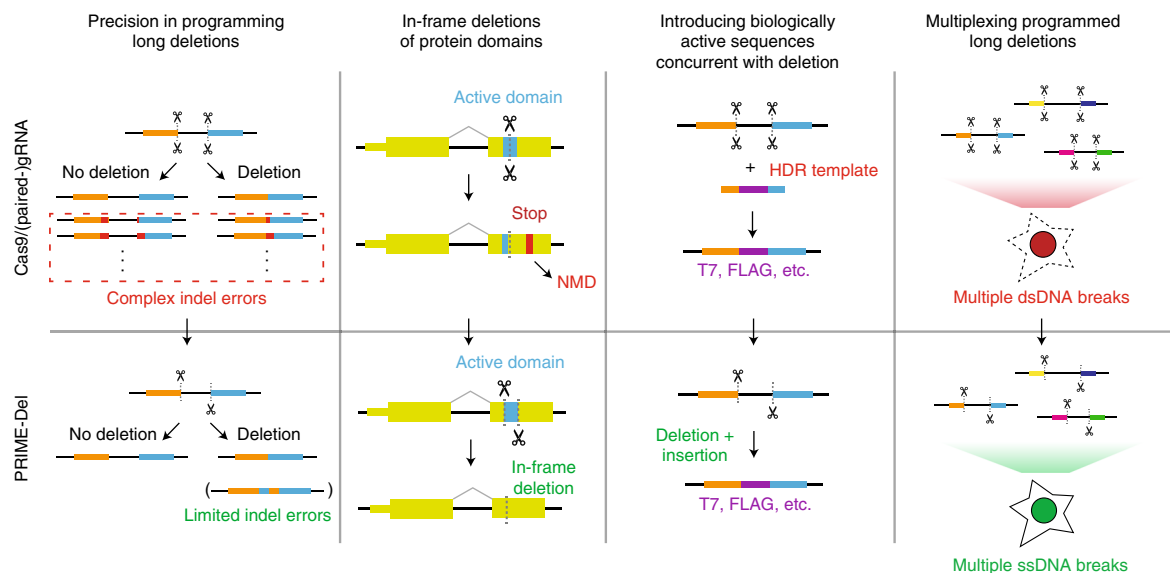


Fig. 5 | Potential advantages of using PRIME-Del in various genome editing applications. The PRIME-Del strategy can be used to program precise genomic deletions without generation of short indel errors at Cas9 target sequences. Precision deletion, combined with the ability to insert a short arbitrary sequence at the deletion junction, may allow robust gene knockout of active protein domains without generating a premature in-frame stop codon, which can trigger the nonsense-mediated decay pathway. PRIME-Del may also allow replacement of genomic regions up to 10 kb with arbitrary sequences, such as epitope tags or RNA transcription start sites. Single-stranded (ss) breaks generated during PRIME-Del are likely to be less toxic to the cell, especially when multiple regions are edited in parallel, potentially facilitating its multiplexing.

inducing shorter deletions. For the 1-kb deletion, both PRIME-Del and the Cas9/paired-sgRNA method achieved almost 3% deletion efficiency. For the 10-kb deletion, PRIME-Del and the Cas9/paired-sgRNA method achieved 0.8% and 1.6% deletion efficiency, respectively. On sequencing amplicons derived from a PCR specific to the post-deletion junction, 98% and 97% of reads lacked indel errors at the junction with PRIME-Del for the 1-kb and 10-kb deletions, respectively, whereas only 47% and 42% of reads lacked indel errors with the Cas9/paired-sgRNA strategy (Fig. 4e).

To test whether the PRIME-Del can be 'multiplexed', we pooled plasmids encoding paired-pegRNAs programming four different but overlapping deletions (118, 252, 469 and 1,064 bp) at the *HPRT1* locus, and transfected HEK293T cells with these together with a plasmid encoding the Prime Editor-2 enzyme. After incubating cells for 4 d and extracting genomic DNA, we used sequencing-based quantification to estimate 5.1%, 8.5% and 2.8% efficiencies for the 118-, 252- and 469-bp deletions, respectively, and ddPCR to estimate 2% efficiency for the 1,064-bp deletion (Supplementary Fig. 6). Altogether, we estimate that 18% of *HPRT1* loci carry one of the four programmed deletions, which is comparable to the averaged efficiency of four deletions performed by transfecting a single construct of paired-pegRNA plasmid separately (12%). Our result suggests that PRIME-Del can be used to concurrently program multiple deletions by using pooled paired-pegRNA constructs similar to Cas9/paired-sgRNA method^{5,10,11}.

Extending editing time enhances prime editing efficiency. In contrast with Cas9-mediated DSBs followed by NHEJ, both prime editing and PRIME-Del have high editing precision, producing an intended edit or conserving the original editable sequence. We reasoned that, if the editing efficiencies of prime editing and PRIME-Del are limited by the transient availability of PE2/pegRNA molecules in the cell, extending Prime Editor-2 enzyme and pegRNA expression through stable genomic integration or, alternatively, repetitive transfection would boost the rates of successful editing over time, particularly if uneditable 'dead ends' outcomes are not concurrently accruing.

To facilitate prolonged expression, we generated monoclonal HEK293T and K562 cell lines expressing Prime Editor-2 enzyme

(termed HEK293T(PE2) and K562(PE2), respectively), and transduced them with lentiviral vectors bearing pegRNAs (Supplementary Fig. 7a). We tested two different deletions at *HPRT1* using PRIME-Del (the aforementioned 118-bp and 252-bp deletions at exon 1), along with standard prime editing to insert 3 bp (CTT) into the synthetic HEK3-target sequence¹⁴. In K562(PE2), we observed a steady increase of the correctly edited population over time, both for CTT insertion using prime editing and for the 118- or 252-bp deletion using PRIME-Del. The end-point prime editing efficiencies for the CTT insertion were very high, reaching 90% of targets with correct edits by 19 d after the first transduction of pegRNA into K562(PE2) cells (Supplementary Fig. 7b). The rate of precise deletions using PRIME-Del also reached nearly 50% and 25% for the 118-bp and 252-bp deletions, respectively, by 19 d. In HEK293T(PE2) cells, we observed lower CTT-insertion efficiencies for the first 10 d, but they eventually reached 80–90% by day 19 (Supplementary Fig. 7c). Unexpectedly, we observed the near absence of PRIME-Del-induced deletions in HEK293T(PE2) cells (Supplementary Fig. 7c). Although we cannot rule out cell-type-specific differences in prime editing, we also suspect that the expression level of Prime Editor-2 enzyme and pegRNAs heavily affects the editing efficiency, because subsequent attempts in HEK293T(PE2) cells have resulted in accumulating deletions over time (Supplementary Fig. 7d–f). Together, our results confirm that extended expressions of prime editing or PRIME-Del components can boost efficiency, although it may induce greater off-target effects of prime editing.

Potential applications of PRIME-Del. In the present study, we introduced PRIME-Del, a paired-pegRNA strategy for prime editing, and demonstrated that it achieves high precision for programming deletions, both with and without short programmed insertions. We tested deletions ranging from 20 bp to ~10,000 bp in length at episomal, synthetic genomic and native genomic loci. The editing efficiency on native genes ranged from 1% to 30% with a single round of transient transfection in HEK293T cells, although we also observed that prolonged, high expression of prime editing or PRIME-Del components enhanced editing efficiency in K562

cells. For 12 deletions at 7 genomic loci targeted with PRIME-Del, we observed high precision of editing except at *HPRT1* exon 1, where long insertions were sometimes observed at the deletion junction (~5% of total reads). The GC-rich ends of 3'-DNA flap sequences of the pegRNA pairs used at *HPRT1* exon 1 appear to underlie the long insertions. By optimizing pegRNA design we may be able to eliminate this error mode, and show that lengthening of homology arms tends to decrease the frequency of long insertion errors. To facilitate avoidance of this particular error mode, we have developed an accompanying Python-based webtool for designing PRIME-Del paired-pegRNA sequences, which notifies the user if such sequences are present in designed pegRNA pairs.

However, even with these insertion errors, PRIME-Del consistently demonstrated higher precision than the Cas9/paired-sgRNA strategy, that is, for all 12 genomic deletions tested here, PRIME-Del resulted in fewer erroneous outcomes. For these same twelve cases, PRIME-Del exhibited markedly higher precise-deletion efficiencies for five (greater than a factor of two), comparable efficiencies for five (within a factor of two) and markedly lower efficiencies for two (less than half) compared with the Cas9/paired-sgRNA method. Overall, these observations support the view that PRIME-Del achieves higher precision than the Cas9/paired-sgRNA method without compromising editing efficiency.

A potential design-related limitation of PRIME-Del is that, relative to the conventional Cas9/paired-sgRNA strategy, it constrains the useable pairs of genomic protospacers, because they need to occur on opposing strands with the PAM sequences oriented toward each other (Fig. 1c). However, the development and optimization of an almost PAM-less²² prime editing enzyme²³ would relax this constraint. A further limitation is that, because of their longer length, cloning a pair of pegRNAs in tandem is more challenging than cloning sgRNA pairs. Each pegRNA used in the present study is 135–140 bp in length, such that synthesizing their unique components in tandem as a single, long oligonucleotide approaches the limits of conventional DNA synthesis technology, particularly for goals requiring array-based synthesis of paired pegRNA libraries.

Notwithstanding these limitations, PRIME-Del offers notable advantages over alternatives across several potential areas of application (Fig. 5). Most straightforwardly, PRIME-Del can be used for precise programming of deletions up to 10 kb; we have yet to attempt deletions longer than 10 kb. In addition to the much lower indel error rate observed at the deletion junction compared with the Cas9/paired-sgRNA strategy, inducing paired nicks is less likely to result in large, unintended deletions locally, rearrangements genome wide (chromothripsis)²⁴ or off-target editing^{7,14,25–27}. These characteristics are advantageous for developing therapeutic approaches, for example, where PRIME-Del deletes pathogenic regions such as CGG-repeat expansions in the 5'-UTR of *FMRI*, with no undesired perturbation of nearby or distant sequences^{12,13}.

PRIME-Del also allows simultaneous insertion of short sequences at the programmed deletion junction without substantially compromising its efficiency or precision. Inserting short sequences allows for precise deletions of protein domains while preserving the native reading frame, that is, avoiding a premature stop codon that might otherwise elicit a complex nonsense-mediated decay response^{28,29}. Furthermore, inserting biologically active sequences on deletion is likely to be advantageous in coupling PRIME-Del with technologies, that is, by inserting epitope tags or T7 promoter sequences that can be used as molecular handles within edited genomic loci.

We also expect less toxicity via DNA damage by prime editing-based PRIME-Del than with the conventional Cas9/paired-sgRNA strategy, which may facilitate multiplexing of programmed genomic deletions for frameworks such as scanDel and crisprQTL^{5,6}. For studying the noncoding elements in transcription, efficient and precise deletions up to ~10 kb complements the current

use of deactivated Cas9–tethered KRAB domain for CRISPR–interference, which cannot control the range of epigenetic modifications around target regions. As such, we anticipate that PRIME-Del could be broadly applied in massively parallel functional assays to characterize native genetic elements at base-pair resolution.

Online content

Any methods, additional references, Nature Research reporting summaries, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-021-01025-z>.

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Methods

The pegRNA/sgRNA design. For pegRNA/sgRNA design, we initially used CRISPOR³⁰ to select for 20-bp CRISPR–Cas9 spacers within a given region of interest. We avoided spacers annotated as inefficient, including U6/H1 terminator and GC-rich sequences, and generally selected spacers that had higher predicted efficiencies (Doench scores for U6-transcribed sgRNAs³¹). The length of the RT-template portion of a pegRNA was initially set to 30 bp and extended by 1–2 bp if it ended in G or C^{14,32}.

Webtool for PRIME-Del paired-pegRNA design. To facilitate PRIME-Del paired-pegRNA design, we developed a Python-based webtool that automates the design process. The software takes a FASTA-formatted sequence file as the input, identifies all possible PAM sequences within the provided region and initially generates all potential paired pegRNA sequences to program deletions. The software can also optionally take as input scored sgRNA files generated using FlashFry³³, CRISPOR³⁰ or GPP sgRNA designer³⁰; this is highly recommended to identify effective CRISPR–Cas9 spacers. For FlashFry and CRISPOR, sgRNA spacers with MIT specificity scores³⁴ <50 are filtered out as recommended by CRISPOR. From initially generated pegRNA pairs, the software selects relevant ones based on additional user-provided design parameters. For example, the user can define the deletion size range. The user can also define the start and end position of desired deletion, and the software will filter to pegRNA pairs present in windows centered at those coordinates. The pegRNAs for deletions with junctions that do not fall at PAM sites can be designed using the option ‘--precise’ (-p), which adds insertion sequences to both pegRNAs to facilitate the desired edit.

The PRIME-Del design software also enables additional design constraints to be specified. The pegRNA RT-template length (also known as the homology arm) is set to 30 bp by default, unless specified otherwise by the user. The pegRNA primer-binding sequence (PBS) length is set to 13 bp from the PE2 nick site by default, unless specified otherwise by the user. The nick position relative to the PAM sequence is predicted using previously identified parameters³⁵, and the RT-template length is adjusted accordingly if the predicted likelihood of generating a nick at a noncanonical position is >25%. PegRNA sequences that include RNA polymerase III terminator sequences (more than four consecutive Ts) are filtered out. The software generates warning messages if more than 4 or 5 bp in either 3′-DNA-flap are either G or C. Code is available at <https://github.com/shendurelab/Prime-del> and an interactive webpage is available at <https://primedel.ucr.appspot.com>.

PegRNA cloning. After designing pegRNA pairs, we followed the Golden-Gate cloning strategy outlined by Anzalone et al.¹⁴, assembling three double-stranded (ds)DNA fragments and one plasmid backbone. The first dsDNA fragment contains the pegRNA-1 spacer sequence, annealed from two complementary synthetic single-stranded DNA oligonucleotides (Integrated DNA Technologies (IDT)) with 4-bp 5′-overhangs. The second dsDNA fragment contains the pegRNA-1 sgRNA scaffold sequence, annealed from two DNA oligonucleotides with 5′-end phosphorylation at the end of a 4-bp overhang. The third dsDNA fragment contains the pegRNA-1 RT-template sequence and PBS, pegRNA-1 terminator sequence (six consecutive Ts) and pegRNA-2 sequence with H1 promoter sequence. This was generated by appending the pegRNA-1 and pegRNA-2 portions to two ends of gene fragments (purchased as gBlocks from IDT) by PCR amplification. The gene fragments contained the pegRNA-1 terminator sequence, H1 promoter sequence, pegRNA-2 spacer sequence and pegRNA-2 sgRNA scaffold sequences. The forward primer included the BsmBI or BsaI restriction site, pegRNA-1 RT-template sequence and PBS. The reverse primer included the pegRNA-2 RT-template, PBS, and BsmBI or BsaI restriction site. PCR fragments (sized between 300 and 400 bp) were purified using 1.0× AMPure (Beckman Coulter) and mixed with two other dsDNA fragments and linearized backbone vector, with corresponding overhangs for Golden-Gate-based assembly mix (BsmBI or BsaI Golden-Gate assembly mix from New England Biolabs). For the pegRNA cloning backbone, we used either the GG-acceptor plasmid (Addgene, catalog no. 132777) or piggyBAC-cargo vector that carries the blasticidin-resistance gene. Each construct plasmid was transformed into Stbl Competent *Escherichia coli* (New England Biolabs, catalog no. C3040H) for amplification and purified using a miniprep kit (QIAGEN). Cloning was verified using Sanger sequencing (Genewiz).

Tissue culture, transfection, lentiviral transduction and monoclonal line generation. HEK293T and K562 cells were purchased from American Type Culture Collection. HEK293T cells were cultured in Dulbecco's modified Eagle's medium with high glucose (Gibco), supplemented with 10% fetal bovine serum (Rocky Mountain Biologicals) and 1% penicillin–streptomycin (Gibco). K562 cells were cultured in RPMI 1640 with L-glutamine (Gibco), supplemented with 10% fetal bovine serum (Rocky Mountain Biologicals) and 1% penicillin–streptomycin (Gibco). HEK293T and K562 cells were grown with 5% CO₂ at 37°C.

For transient transfection, about 50,000 cells were seeded to each well in a 24-well plate and cultured to 70–90% confluence. For prime editing, 375 ng of Prime Editor-2 enzyme plasmid (Addgene, catalog no. 132775) and 125 ng of pegRNA or paired-pegRNA plasmid were mixed and prepared with transfection reagent (Lipofectamine 3000) following the recommended protocol from the

manufacturer. For deletion using Cas9/paired-sgRNA, 375 ng of Cas9 plasmid (Addgene, catalog no. 52962) was used instead of Prime Editor-2 enzyme plasmid. Cells were cultured for 4–5 d after the initial transfection, unless noted otherwise, and its genomic DNA was harvested either using DNeasy Blood and Tissue kit (QIAGEN) or following the cell lysis and protease protocol from Anzalone et al.¹⁴.

For lentiviral generation, about 300,000 cells were seeded to each well in a 6-well plate and cultured to 70–90% confluence. Lentiviral plasmid was transfected along with the ViraPower lentiviral expression system (Thermo Fisher Scientific) following the recommended protocol from the manufacturer. Lentivirus was harvested following the same protocol, concentrated overnight using Peg-it Virus Precipitation Solution (SBI), and used within 1–2 d to transduce either K562 or HEK293T cells without a freeze–thaw cycle.

For transposase integration, 500 ng of cargo plasmid and 100 ng of Super piggyBAC transposase expression vector (SBI) were mixed and prepared with transfection reagent (Lipofectamine 3000) following the manufacturer's recommended protocol. Prime Editor-2, enzyme-expressing, single-cell clones were generated by integrating PE2 using the piggyBAC transposase system, selected by marker (puromycin-resistance gene), single cell sorted into 96-well plates using flow-sort apparatus, cultured for 2–3 weeks until confluence and screened for PE activity by transfecting CTT-inserting pegRNA alone (Addgene, catalog no. 132778) and sequencing the HEK3-target loci.

DNA-sequencing library preparation. To quantify programmed deletion efficiency and possible errors generated by PRIME-Del, we amplified the targeted region from purified DNA (~200 to ~1,000 bp in length) using two-step PCR and sequenced using an Illumina sequencing platform (NextSeq or MiSeq; Supplementary Fig. 1a). Each purified DNA sample contains wild-type and edited DNA molecules, which were amplified together using the same pairs of primers through each PCR reaction. For the PCR amplification, we designed a pair of primers for each genomic locus (amplicon) where entire amplicon sizes, with or without deletion, were >200 bp to avoid potential problems in PCR amplification, purifying of PCR products and clustering onto the sequencing flow cell.

The first PCR reaction (KAPA Robust) included 300 ng of purified genomic DNA or 2 µl of cell lysate, and 0.04–0.4 µM of forward and reverse primers in a final reaction volume of 50 µl. We programmed the first PCR reaction to be: (1) 3 min at 95°C, (2) 15 s at 95°C, (3) 10 s at 65°C, (4) 45 s at 72°C, 25–28 cycles of repeating steps 2–4, and (5) 1 min at 72°C. Primers included sequencing adapters to their 3′-ends, appending them to both termini of PCR products that amplified genomic DNA. After the first PCR step, products were assessed on 6% Tris–borate–ethylenediaminetetraacetic acid gel and purified using 1.0× AMPure (Beckman Coulter), and added to the second PCR reaction that appended dual sample indices and flow cell adapters. The second PCR reaction program was identical to the first PCR program except we ran five to ten cycles. Products were again purified using AMPure and assessed on the TapeStation (Agilent) before being denatured for the sequencing run. For long deletions that generate amplicons sized 200–300 bp, we used a MiSeq sequencing platform at low (8 pM) input DNA concentration to minimize the short amplicons replacing the long amplicons during clustering, aiming at a cluster density of 300,000–400,000 per mm². Denatured libraries were sequenced using either Illumina NextSeq or MiSeq instruments following the manufacturer's protocols.

For appending 15-bp UMIs, we performed the first PCR reaction in two-steps: first, genomic DNA was linearly amplified in the presence of 0.04–0.4 µM of single forward primer in two PCR cycles using KAPA Robust polymerase. We programmed the UMI-appending linear PCR reaction to be: (1) 3 min and 15 s at 95°C, (2) 1 min at 65°C, (3) 2 min at 72°C, 5 cycles of repeating steps 2 and 3, (4) 15 s at 95°C, (5) 1 min at 65°C, and (6) 2 min at 72°C and another 5 cycles of repeating steps 5 and 6. This reaction was cleaned up using 1.5× AMPure, and subject to the second PCR with forward and reverse primers. In this case, the forward primer anneals to the upstream of the UMI sequence and is not specific to the genomic loci. After PCR amplification, products were cleaned up and added to another PCR reaction that appended dual sample indices and flow cell adapters, similar to other samples.

Sequencing data processing and analysis. We designed the sequencing layout to cover at least 50 bp away from the deletion junction in each direction (Supplementary Fig. 1a). In case of the paired-end sequencing, PEAR³⁶ was used to merge the paired-end reads with default parameters and ‘-e’ flag to disable the empirical base frequencies. When a 15-bp UMI was present in the sequencing reads, we used a customized Python script to find all reads that shared the same UMI, and collapsed them into a single read with the most frequent sequence. The resulting sequencing reads were aligned to two reference sequences (with or without deletion) generally using the CRISPResso2 software³⁷. Default alignment parameters were used in CRISPResso2, with the gap-open penalty of –20, the gap-extension penalty of –2 and the gap incentive value of 1 for inserting indels at the cut/nick sites. The minimum homology score for a read alignment was explored between 50 and 95 for different amplicon lengths. Customized Python and R scripts were used to analyze the alignment results from CRISPResso2.

Alignment was done using two reference sequences (wild-type and deletion) of the same sequence length, generating two sets of reads with respective reference

sequences. Deletion efficiencies were calculated as the fraction of the total number of reads aligning to the reference sequence with deletion over the total number of reads aligning to either references. Genome editing has three types of error modes: substitution, insertion and deletion. Each error frequency was plotted across two reference sequences, highlighting in each such plot the Cas9(H840A) nick site and the 3'-DNA flap incorporation sites.

The ddPCR assay. We designed ddPCR probes following the recommended parameters by BioRad Laboratories. We purchased pre-mixed reference probes and primers for the *RPP30* gene from BioRad Laboratories. Probes and PCR primers were purchased from IDT. Probes were modified with FAM on their 5'-ends and included double quenchers (IDT PrimeTime quantitative PCR probes). Probe sequences were specifically designed to cover the deletion junction for detecting precise deletion products³⁴. For detecting each deletion, we prepared a 20× primer mix made up of 18 μM forward primer, 18 μM reverse primer and 5 μM FAM-labeled probe in 50 mM Tris-HCl buffer, pH 8.0 (at room temperature). Then 25 μl of ddPCR reaction mixes was made up of 12.5 μl of 2× Supermix for Probes (no dUTP; BioRad Laboratories), 1.25 μl of 20× HEX-modified *RPP30* reference mix (BioRad Laboratories), 1.25 μl of 20× FAM-modified primer mix, 0.5 μl of cell lysate containing genomic DNA and 9.5 μl of DNase-free water. We added 20 μl of ddPCR reaction mix to 70 μl of droplet generation oil for probes and used a QX200 Droplet generator (BioRad Laboratories) to generate droplets. Droplets were transferred to ddPCR 96-well plates (BioRad Laboratories) and run on 96-well thermocyclers (Eppendorf) with the following program: (1) 10 min at 95°C, (2) 30 s at 94°C, (3) 1 min at 50°C, 41 cycles of repeating steps 2 and 3, (4) 10 min on 98°C and (5) cooled down to 4°C before loading to a QX200 Droplet reader. Temperature ramps were limited to 1°C per s on all steps on thermocyclers. We used the QX200 Droplet reader and BioRad QuantaSoft Pro software to visualize and analyze ddPCR experiments. The deletion efficiencies were taken from the ratio of FAM⁺ (precise-deletion) to HEX⁺ (*RPP30* reference for genomic DNA loading) events.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Raw sequencing data have been uploaded on the Sequencing Read Archive and made available to the public with associated BioProject accession no. PRJNA692623. Selected plasmids used for programming genomic deletions are available from Addgene (catalog nos. 172655, 172656, 172657 and 172658).

Code availability

Source code for PRIME-Del is available at <https://github.com/shendurelab/Prime-del>. An interactive webpage for designing pegRNAs for PRIME-Del is available at <https://primedel.ucr.appspot.com>.

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Author contributions

J.C., C.C.S. and J.S. conceived the project. J.C. designed and performed experiments with guidance from W.C. and J.S., and assistance from W.C., C.C.S., C.L., F.M.C., A.L., R.M.D. and B.M. F.M.C. and W.Y. contributed to validation data. J.C., W.C. and J.S. analyzed the data. W.C. developed the software included in the manuscript. J.C. and J.S. wrote the manuscript with input from the other authors.

Competing interests

The University of Washington has filed a patent application based on this work, in which J.C., W.C. and J.S. are listed as inventors. The remaining authors declare no competing interests.

Additional information

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Software and code

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Data collection

We used Illumina next-generation sequencing platform (Nextseq 550 and Miseq) and associated software within the instruments for data collection.

Data analysis

We have used BCL2fastq (version 2.20) and CRISPResso2 for data analysis. We have deposited prime editing gRNA design software at GitHub (<https://github.com/shendurelab/Prime-del>)

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Sample size	No sample-size calculation was performed. Values (editing efficiencies) measured in technical replications were used to calculate mean and standard deviations.
Data exclusions	No data was excluded from the analysis.
Replication	All genome editing attempts were performed at least with three technical replicates (separate transfections or transduction of cells cultured separately previous to transfections or transductions).
Randomization	This is not relevant to our study, where measurements were taken on the human cells cultured in vitro.
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